Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 03-Apr-2026

---

1. Administrative & Study Identification

Full Study Title: A Phase II Study of Dasatinib in Children and Adolescents With Newly Diagnosed Chronic Phase Chronic Myelogenous Leukemia (CML) or With Ph+ Leukemias Resistant or Intolerant to Imatinib **Short Title/Acronym:** CA180-226 (Phase 2 Dasatinib in Pediatric CML) **Protocol Number:** CA180-226 **NCT Number:** NCT00777036 **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** Dasatinib (Sprycel®) | **ATC Code:** L01EA02 **Phase of Development:** Phase II **Medical Monitor:** Bristol Myers Squibb Clinical Operations / Medical Monitor

---

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of dasatinib in pediatric patients by determining the major cytogenetic response (MCyR) rate in patients with imatinib-resistant/intolerant CML-CP, the complete hematologic response (CHR) rate in patients with advanced phase disease, and the complete cytogenetic response (CCyR) rate in newly diagnosed CML-CP patients. **Secondary Objectives:** To evaluate the time to Major Molecular Response (MMR), Event-Free Survival (EFS), Progression-Free Survival (PFS), overall survival (OS), safety, and the pharmacokinetics of dasatinib tablets and powder for oral suspension (PFOS) in pediatric cohorts.

2.2 Background & Rationale To address the gap in treatment options and formulations for pediatric patients with Philadelphia chromosome-positive (Ph+) leukemias. Children with Chronic Myeloid Leukemia (CML) often present with more aggressive disease characteristics than adults. Prior to this study, there was a critical unmet need for safe, effective, second-generation Tyrosine Kinase Inhibitor (TKI) therapies for pediatric patients who are either newly diagnosed or resistant/intolerant to first-line imatinib. Furthermore, evaluating a powder for oral suspension (PFOS) provides a vital dosing alternative for young children unable to swallow tablets.

---

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Non-randomized cohorts) **Blinding:** Open-label **Randomization:** N/A (Assigned to cohorts based on disease phase and prior treatment status) **Trial Status:** COMPLETED

3.2 Planned Enrollment & Duration Target N: 133 pediatric patients **Number of Sites:** Multicenter global clinical trial (approx. 26-50+ locations) **Estimated Study Start:** November 2008 **Estimated Primary Completion:** May 2018 (Final long-term follow-up analyses)

---

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age < 18 years (originally < 21 years; amended during study). 2. Confirmed diagnosis assigned to one of three cohorts: - **Cohort 1:** CP-CML who prove resistant or intolerant to imatinib. - **Cohort 2:** Ph+ ALL, AP-CML, or BP-CML who are resistant or intolerant to or who relapse after imatinib therapy. - **Cohort 3:** Newly diagnosed, treatment naive CP-CML. 3. Lansky or Karnofsky performance status > 50. 4. Life expectancy ≥ 12 weeks. 5. Adequate hepatic and renal function. 6. Written informed consent obtained from the subject, or from parents/legal guardians for minor subjects. 7. **PK Substudy Target Population:** Must have CP-CML and be taking daily dasatinib (tablets or PFOS) tolerating a dose of at least 60 mg/m^2 (tablet) or 72 mg/m^2 (PFOS). Prior exposure to imatinib or other TKI therapy is permissible, and participants must obtain specific written informed consent for the substudy.

4.2 Exclusion Criteria 1. Eligibility for potentially-curative therapy including hematopoietic stem-cell transplantation. 2. Symptomatic CNS involvement (other than signs and symptoms caused by leptomeningeal disease). 3. Isolated extramedullary disease. 4. Prior therapy with Dasatinib. 5. Presence of significant cardiovascular disease, prolonged QTc, or history of severe pleural/pericardial effusion. 6. **PK Substudy specific:** Subjects are not allowed to use proton pump inhibitors, H2 antagonists, CYP3A4 inhibitors, and inducers when entering the PK substudy.

---

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Dasatinib 60 mg/m^2 once daily (for CML-CP) or 80 mg/m^2 once daily (for advanced phase CML/ALL). Doses were administered as intact tablets or mathematically adjusted to 72 mg/m^2 once daily for the Powder for Oral Suspension (PFOS) to match tablet exposure. Dosages were recalculated every 3 months based on body surface area (BSA) and weight. **Route of Administration:** Oral **Duration of Treatment:** Continued until disease progression, unacceptable toxicity, or patient/physician preference.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care; hematopoietic growth factors (e.g., G-CSF) for the management of resistant myelosuppression. **Prohibited:** Concomitant use of strong CYP3A4 inhibitors or inducers (dose interruption/modification required if unavoidable); St. John's wort; concurrent medications known to prolong the QTc interval were to be used with extreme caution. (PPIs, H2 antagonists, CYP3A4 inhibitors/inducers explicitly excluded for PK substudy).

---

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Cytogenetic/Molecular Labs, ECG
Baseline	Day 0	Treatment Initiation, Pharmacokinetic (PK) sampling (for sub-study)
Interim	Months 3, 6, 12, 24	Safety Labs, Bone Marrow Aspirate/Cytogenetics, PCR for BCR-ABL1, Bone Growth Monitoring
End of Study	Follow-up	Final physical exam, Safety follow-up, Survival and Progression tracking

---

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Major Cytogenetic Response (MCyR) Rate:** Evaluated in Cohort 1. Defined as the proportion of treated participants achieving complete (0%) or partial (1%-35% Ph+ metaphases) response in bone marrow. 2. **Complete Hematologic Response (CHR) Rate:** Evaluated in Cohort 2. Defined as achieving no more than 5% blasts in bone marrow and normal WBC count without blasts in peripheral blood. 3. **Complete Cytogenetic Response (CCyR) Rate:** Evaluated in Cohort 3. Defined as 0% Ph+ metaphases in at least 20 metaphases in bone marrow.

7.2 Secondary & Exploratory Endpoints 1. Major Cytogenetic Response (MCyR) Rate in Cohort 2. 2. Complete Hematologic Response (CHR) Rate in Cohorts 1 and 3. 3. Rate of Best Cytogenetic Response. 4. Time to Major Molecular Response (MMR) and Cumulative MMR rate by 12 and 24 months. 5. Progression-Free Survival (PFS) and Event-Free Survival (EFS) evaluated at 24 and 48 months.

---

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection focusing on TKI-associated toxicities, specifically myelosuppression, bleeding events, bone growth/development tracking, and fluid retention. **Serious Adverse Events (SAEs):** Standard 24-hour reporting. Highly monitored events included Grade 3/4 cytopenias, gastrointestinal hemorrhage, and potential pulmonary arterial hypertension (PAH) or pleural/pericardial effusion. **Data Monitoring Committee (DMC):** Independent data monitoring committee oversaw the safety and efficacy results for the pediatric populations.

---

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Treated Population (all patients who received at least one dose of dasatinib) for safety and primary efficacy evaluations. Confidence intervals for primary response rates were calculated by the Clopper-Pearson exact method. **Sample Size Justification:** The study was designed to be descriptive and estimation-based. The sample sizes per cohort were deemed sufficient to construct 95% Confidence Intervals around the target threshold response rates. **Handling of Missing Data:** Evaluated utilizing an intent-to-treat framework; patients who discontinued the study prior to achieving a response were considered non-responders in the cumulative response rate calculations.

---

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard onsite and remote clinical monitoring by the sponsor to ensure protocol adherence, accurate drug dispensation (tablets vs. PFOS), and data integrity. **Audit Trail:** Continuous tracking via 21 CFR Part 11 compliant eCRF systems and central laboratory data reconciliation.

---

11. Study Identifiers & Governance

Primary ID: CA180-226 **Registry Number:** NCT00777036 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Phase II Study of Dasatinib in Children and Adolescents With Newly Diagnosed Chronic Phase Chronic Myelogenous Leukemia (CML) or With Ph+ Leukemias Resistant or Intolerant to Imatinib **Official Regulatory Title:** A Phase II Study of Dasatinib Therapy in Children and Adolescents With Newly Diagnosed Chronic Phase Chronic Myelogenous Leukemia or With Ph+ Leukemias Resistant or Intolerant to Imatinib

---

12. Clinical Framework (Leukemia Specific)

Disease Area: Leukemia **Primary Condition:** Chronic Myelogenous Leukemia (CML) and Ph+ Leukemias in Pediatric Patients **Diagnosis Threshold:** Philadelphia chromosome-positive (Ph+) / BCR-ABL1 transcript confirmation **Medical Methodology:** Tyrosine Kinase Inhibitor (TKI) Therapy **Optimization Target:** Complete Cytogenetic Response (CCyR) and Major Molecular Response (MMR)

---

13. Study Procedures & Methodology

Management Pathway: Pediatric CML standard clinical management pathway **Education Protocol (HCP):** TKI safety monitoring (focused on bone growth and myelosuppression mitigation) **Education Protocol (Patient):** Daily oral adherence training (specifically managing transition and compliance between PFOS formulation and tablets) **Quality Audit Mechanism:** Central laboratory verification for BCR-ABL1 transcript levels and routine growth velocity tracking

---

14. Observation & Follow-Up Schedule

Total Duration: Minimum 48 to 96+ weeks (long-term follow-up often extending beyond 4 years) **Onsite Visit Cadence:** Baseline, Months 3, 6, 12, 24, and every 6-12 months thereafter **Remote Monitoring Frequency:** Routine safety calls and lab verifications between clinical clinic visits **Checklist Review Interval:** Quarterly (Q12W) recalculations of Body Surface Area (BSA) / weight for precise dosing

---

15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] | ___ |** ***[DD-MMM-YYYY] | | Lead Statistician | [Name] |*** _____ | [DD-MMM-YYYY] |